MAT A 31

Week 4: *NOTES*

Textbook: Taalman, Kohn: CALCULUS, *Single Variable.*

DEFINITION 1 **Algebraic Definition of Limit**

$\lim_{x \to C} f(x) = L$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that

if $0 < |x - c| < \delta$, then $|f(x) - L| < \varepsilon$

Equivalence of Geometric and Algebraic Definitions of Limit:

Given: $c = 3,\quad L = 5,\quad \varepsilon = 0.1,\quad f(x) = 2x - 1$

Find: δ

Let's prove the statement $\lim_{x \to 2}(7 - x) = 5$ using the Algebraic Definition of Limit.

Steps in $\delta - \varepsilon$ proofs of $\lim_{x \to C} f(x) = L$

1. Let $\varepsilon > 0$ be an arbitrary positive number.
2. Find δ in terms of ε (establish connection between δ and ε)
3. Show that if values of x in the interval $0 < |x - c| < \delta$ then $|f(x) - L| < \varepsilon$.

EXAMPLES

1. Prove that $\lim_{x \to 3}(4x - 5) = 7$.

2. Prove that $\lim_{x \to 2^+} 3\sqrt{2x-4} = 0$

3. Prove that $\lim_{x \to -2^-} \frac{1}{x+2} = -\infty$

4. For the $\lim_{x \to 5} \sqrt{x-1} = 2$, find a $\delta > 0$ that works for $\varepsilon = 1$

An open interval of radius 3 about $c = 5$
will lie inside the open interval (2,10)

The process of finding a $\delta > 0$ for a given $f(x), L, c$ *and* $\varepsilon > 0$ such that

if values of x is in the interval $0 < |c-5| < \delta$ *then* $|f(x) - L| < \varepsilon$:

<u>Step1.</u> *Solve the inequality* $|f(x) - L| < \varepsilon$ to find an open interval (*a,b*), containing *c* on which the inequality holds for all $x \neq c$.
Step 2. *Find a value of* $\delta > 0$ that places the open interval $(c - \delta, c + \delta)$ centered at *c* inside the interval (*a,b*). The inequality $|f(x) - L| < \varepsilon$ will hold for all $x \neq c$ in this δ interval.

5. Prove $\lim_{x \to 1}(x^2 - 6x + 5) = 0$

6. Prove $\lim_{x \to 5}(x^2 - 6x + 7) = 2$

7. Prove $\lim_{x \to 1}(x^2 + 4x - 5) = 0$

8. Prove $\lim_{x \to 4} \sqrt{x} = 2$